Implementation of AI algorithms and UI changes and Enhancing performance marketing and sales ROI

A dissertation submitted in partial Fulfillment of the requirement for the award of Degree in

Bachelor of Fashion Technology (Apparel Production)

Submitted By

Divyansh Srivastava (BFT/20/707)

Under the Guidance of

Dr. Omkar Singh

(Professor, NIFT Patna)

Department of Fashion Technology

National Institute of Fashion Technology, (Patna)

May, 2024

Abstract

The competitive marketing landscape craves innovation to maximize return on investment (ROI). This project unveils the synergistic potential of artificial intelligence (AI) algorithms and strategic user interface (UI) modifications to revolutionize performance marketing strategies.

At the heart of this approach lies the transformative power of AI. By meticulously analysing vast customer data sets and meticulously dissecting behaviour patterns, AI unlocks a treasure trove of insights. This empowers the creation of highly targeted, data-driven marketing campaigns. Imagine tailoring messaging and promotions to individual customer preferences and purchase histories – a level of personalization that traditional methods simply cannot achieve. The result: a dramatic surge in conversion rates and customer engagement, directly translating to increased sales figures.

However, AI's prowess extends beyond mere targeting. Advanced algorithms can predict customer churn, enabling proactive interventions to retain valuable clientele before they abandon ship. Additionally, AI can personalize product recommendations, guiding customers on a seamless journey towards their desired purchases. This translates to a more satisfying customer experience, fostering loyalty and brand advocacy. Satisfied customers become brand evangelists, spreading positive word-of-mouth and organically attracting new customers – a win-win for businesses.

Complementing AI's power is the strategic manipulation of the user interface (UI). Streamlining the customer journey is paramount. By simplifying navigation, implementing intuitive search functions, and highlighting relevant products or services, UI enhancements ensure customers can effortlessly find what they seek. This not only boosts conversion rates but also increases customer satisfaction. Imagine an e-commerce store where AI-powered product recommendations anticipate your needs, and a frictionless checkout process allows for effortless transactions. Frustration is minimized, replaced by a sense of delight, driving repeat business and positive online reviews.

While AI offers immense potential, a poorly designed recommendation engine can backfire spectacularly. Imagine a website persistently suggesting irrelevant products or bombarding users with generic recommendations. This can lead to:

- **Frustration and Disengagement:** Users bombarded with irrelevant suggestions quickly lose interest. Imagine searching for hiking boots and being flooded with recommendations for swimsuits. This disconnect creates frustration and drives users away.

- **Echo Chambers and Filter Bubbles:** AI algorithms can become trapped in "echo chambers," reinforcing existing user preferences instead of introducing them to new and potentially valuable options. This limits discovery and can lead to stagnant customer journeys.
- **Loss of Trust:** Repeatedly receiving irrelevant recommendations erodes user trust in the brand and the platform's ability to understand their needs. This can lead to users actively seeking out competitors who offer a more personalized experience.

The Synergistic Solution:

By combining the power of AI with strategic UI modifications, this project aims to overcome these pitfalls and unlock peak performance marketing:

- **Data-Driven Personalization:** AI analyzes vast customer data sets and behavior patterns to deliver hyper-personalized recommendations. This ensures users see relevant products that resonate with their interests.
- **Dynamic Content:** UI elements like product carousels and banner ads can be dynamically populated with AI-powered suggestions, maximizing the potential for discovery while minimizing the risk of irrelevant content.
- **User Control & Transparency:** The UI empowers users to refine their recommendations by providing feedback or adjusting preferences. This transparency fosters trust and ensures the system remains user-centric.

The project goes beyond simply implementing these advancements. It meticulously evaluates their effectiveness through a rigorous analysis of key performance indicators (KPIs). Metrics like click-through rates, conversion rates, and overall sales figures will be closely monitored. The project aims to establish a direct correlation between the implementation of AI algorithms and UI changes, and a measurable improvement in performance marketing ROI.

In essence, this project investigates the transformative potential of AI and UI working in unison. By harnessing the power of AI for hyper-personalized targeting, customer journey optimization, and predictive analytics, along with UI modifications that provide a seamless user experience, the project seeks to unlock a new era of performance marketing. This era promises significant improvement in ROI, driving sustainable business growth and fostering a loyal customer base that advocates for your brand

Certification

*This is to certify that this Project Report titled "***Implementation of AI Algorithms and UI Changes for Enhanced Performance Marketing and Sales ROI***" is based on my,* **Divyansh Srivastava***'s original research work, conducted under the guidance of Dr. Omkar Singh towards partial fulfillment of the requirement for award of the Bachelor's Degree in Fashion Technology (Apparel Production), of the National Institute of Fashion Technology, Campus. No part of this work has been copied from any other source. Material, wherever borrowed has been duly acknowledged."*

(Divyansh Srivastava)

Signature of Author/Researchers

(*Dr. Omkar Singh ,Professor, NIFT Patna***)**

Signature of Guide

JACK&JONES

Date: - **06th May 2024**

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Divyansh Srivastava** has undergone internship training with our **E-Commerce Department** from **08th Jan. 2024** to **17th April 2024.**

During the above-mentioned training period, we found him to be sincere and hardworking.

We wish her success in him future endeavours.

For Best United India Comforts Pvt. Ltd

(Ruchi Karwar)
Authorized Signatory

BEST UNITED INDIA COMFORTS PRIVATE LIMITED
Registered Office: The Lalit Residency, 2nd Floor, The Lalit Mumbai, Sahar Airport Road, Sahar, Andheri (E), Mumbai- 400 059
Tel. No. 022- 66 48 9000 / 61747600
(CIN: U52390MH2009PTC190686)

Acknowledgements

I would like to express my heartfelt gratitude to all the individuals who have played a significant role in the successful completion of this project. Their unwavering support, guidance, and expertise have been instrumental in the achievement of the goals.

First and foremost, I extend my deepest appreciation to my faculty guide, Dr. Omkar Singh (Professor, NIFT Patna) for his valuable mentorship, constant encouragement, and insightful feedback. I would also like to extend my thanks to NIFT for providing me this opportunity to undergo this Graduation Project.

Furthermore, I would like to express my appreciation to Bestseller for giving me this platform to conduct and complete my project. Last but not least, I extend my thanks to all the individuals from Bestseller and also everyone who participated in the survey and provided valuable feedback. The contributions given by everyone have been invaluable, and without their support, this project would not have been possible.

Table of Contents

List of Figures

lists of Tables

Chapter 1. All about Bestseller

1. About Bestseller

Bestseller is a leading international clothing company headquartered in Denmark, boasting a rich history and a diverse brand portfolio. This behemoth of the fashion industry has carved a niche for itself through its focus on "fast fashion" and its ability to cater to a wide range of demographics across the globe.

1.1 From Humble Beginnings to Global Recognition

The Bestseller story began in 1975 when Troels Holch Povlsen and Merete Bech Povlsen opened a single store in Ringkøbing, Denmark. Initially, they functioned as an outlet for Troels' uncle's clothing, but their entrepreneurial spirit soon led them to import their own products and expand into wholesale partnerships.

This initial success laid the foundation for Bestseller's remarkable growth. The company embraced a dynamic business model, capitalizing on the fast fashion trend. This approach involves quickly transitioning trendy designs from concept to store shelves, allowing them to cater to ever-evolving consumer preferences.

1.2 A Multi-Branded Powerhouse

Bestseller's success lies not only in its business model but also in its diverse brand portfolio. Each brand caters to a specific customer segment, ensuring a wide reach and market penetration. Here's a glimpse into some of their most prominent brands:

- **JACK & JONES:** A leading menswear brand known for its sharp tailoring, contemporary styles, and focus on high-quality materials.
- **ONLY:** A popular womenswear brand offering trendy and feminine clothing at accessible price points.
- **VERO MODA:** Another prominent womenswear brand specializing in sophisticated and fashion-forward designs for the modern woman.
- **NAME IT:** A vibrant children's clothing brand catering to the needs of youngsters with playful and comfortable designs.
- **Selected Homme:** A brand focused on premium menswear, offering a more elevated and sophisticated aesthetic.

These are just a few examples, and Bestseller continues to expand its brand portfolio, constantly adapting to evolving fashion trends and consumer preferences.

1.3 Bestseller India: A Powerhouse in the Flourishing Indian Fashion Market

Bestseller, a leading global clothing company, recognized the significant potential of the Indian market. This vast and diverse population presented an opportunity for fashion expansion. To capitalize on this, Bestseller established a subsidiary – Bestseller India – specifically designed to cater to Indian tastes and preferences.

1.3.1 Strategic Operations for Market Domination

Bestseller India functions as the Indian arm of the global fashion giant. Their core operations encompass several key functions:

- **Distribution and Retail:** They import, distribute, and sell clothing lines from Bestseller's global brand portfolio within India. This includes popular names like Jack & Jones, Vero Moda, Only, and Selected Homme.
- **Brand Management:** While adhering to the global brand identity, Bestseller India likely tailors marketing strategies and product offerings to resonate with Indian customers and cultural nuances. This ensures a deeper connection with the local audience.
- **Market Expansion:** They play a vital role in identifying new growth opportunities within the Indian market. This could involve opening new stores in untapped regions, establishing partnerships with local retailers, or exploring online marketplaces for wider reach.

1.3.2 A Spectrum of Brands for Diverse Tastes

Bestseller India caters to a wide range of fashion sensibilities through the following prominent brands:

- **Jack & Jones:** A leading menswear brand known for its sharp tailoring, contemporary styles, and focus on high-quality materials.
- **Vero Moda:** A go-to brand for women seeking sophisticated and fashion-forward designs.
- **Only:** This brand offers trendy and feminine clothing at accessible price points for women.
- **Selected Homme:** This brand, also known as Selected in some markets, offers premium menswear with a more elevated and sophisticated aesthetic.

It's important to note that Bestseller owns a diverse brand portfolio globally. While these are the current mainstays in India, Bestseller India might introduce additional brands to the Indian market in the future, catering to even more specific demographics.

1.3.3 Pan-India Presence

Bestseller India boasts a well-established network of stores across the country. Additionally, online retail giants like Amazon.in flipkart.com myntra.com ajio.com carry products from Bestseller brands, offering wider accessibility to Indian customers

1.4 Bestseller's Digital Presence: Websites and Online Shopping Options

While Bestseller itself doesn't have a website for direct clothing sales, they leverage a strategic digital presence to connect with consumers and showcase their brands. Here's a breakdown of their online landscape:

1.4.1 Bestseller Corporate Website

- **URL:** https://bestseller.com/
- **Function:** This website serves as the company's online headquarters. It offers a wealth of information about Bestseller's global operations, sustainability efforts, company philosophy, and the various brands under their umbrella. While it doesn't provide direct access to online shopping, it allows users to explore each brand and potentially find links to their individual websites.

1.4.2 Individual Brand Websites

Each of Bestseller's brands has its own website, acting as a dedicated platform to showcase their unique collections and brand identity. Here are some of the brands available in India, along with their website links:

- **Jack & Jones:** https://www.jackjones.in/ - Offers menswear with a focus on sharp tailoring, contemporary styles, and high-quality materials.
- **Vero Moda:** https://www.veromoda.in/ - Caters to women seeking sophisticated and fashion-forward designs.
- **Only:** https://www.only.in/ - Provides trendy and feminine clothing at accessible price points for women.

- **Selected Homme:** https://www.selected.com/ (Global Website) - Offers premium menswear with a more elevated and sophisticated aesthetic. **Important Note:** While a specific website for Selected Homme India might exist, this global website likely caters to the Indian market as well.

1.4.3 Bestseller India Website (Limited Information):

- **URL:** https://www.bestsellerclothing.in/
- **Function:** This website seems to be focused on legal aspects of Bestseller India's operations. It primarily deals with disclaimers and terms of sale, offering limited information on product lines or online shopping options.

1.4.4 Alternative Shopping Platforms:

Despite the lack of a dedicated e-commerce platform for Bestseller India, their brands are widely available on various online retail platforms in India. Here are a couple of examples:

Amazon India: https://www.amazon.in/ - Carries a wide selection of clothing from various brands, including Bestseller's brands like Jack & Jones, Vero Moda, and Only.

Myntra: https://www.myntra.com/ - Another major online fashion retailer in India that offers products from Bestseller brands.

Tata Cliq: https://www.tatacliq.com/ - A leading Indian online fashion retailer offering a curated selection of brands, including some of Bestseller's brands.

Ajio: https://www.ajio.com/ - An online fashion platform owned by Reliance Retail, which features a selection of Bestseller brands alongside other popular labels.

Flipkart: https://www.flipkart.com/ - A major e-commerce platform in India offering a wide variety of products, including clothing from Bestseller brands.

Nykaa: https://www.nykaa.com/ - A prominent Indian beauty and fashion retailer that also offers a selection of clothing from Bestseller brands, particularly pieces that complement their beauty products.

Chapter 2.0 Understanding the General marketing funnels in the market and our marketing funnels.

2.1 The Sales Funnels in the Online Channel

Ecommerce brands navigate a complex online sales funnel, with potential customers dropping off at various stages. Identifying these drop-off points, or "gaps," is crucial for optimizing marketing strategies and maximizing revenue. While the specific source of sales can vary depending on the brand and industry, some general trends emerge:

1. Organic Search: Often considered the most valuable source of sales, organic search refers to customers finding your website through unpaid search engine results. Strong search engine optimization (SEO) practices ensure your brand appears prominently when users search for relevant products or services.

2. Paid Advertising: Paid search advertising, like Google Ads or social media marketing, allows targeted campaigns to reach a wider audience actively searching for solutions. These platforms offer precise targeting capabilities, enabling you to reach users with high purchase intent.

3. Marketplace Sales: Online marketplaces like Amazon, eBay, or Etsy provide established platforms with significant customer traffic. Brands can leverage these marketplaces to reach a broader audience and benefit from existing customer trust in the platform.

4. Affiliate Marketing: Through partnerships with other websites or influencers, brands can leverage affiliate marketing to reach new audiences. Affiliates promote your products or services in exchange for a commission on sales generated through their unique referral link.

5. Email Marketing: Cultivating a loyal customer base fosters repeat business. Email marketing allows for targeted communication, promoting new products, offering exclusive discounts, and nurturing leads through the sales funnel.

6. Performance Marketing: This results-driven approach focuses on advertising channels where you only pay for specific actions, such as clicks, leads, or completed sales. Examples include pay-per-click (PPC) advertising, affiliate marketing with performance-based payouts, and social media advertising with conversion tracking. Performance marketing allows for highly targeted campaigns and a measurable return on investment (ROI).

2.1.1 Types of Marketing for Brand Growth: Acquisition and Retention

Brands require a multifaceted marketing approach to achieve sustainable growth. Here's a breakdown of key marketing strategies for acquiring new customers and retaining existing ones:

2.1.2 New Customer Acquisition:

- **Brand Awareness Marketing:** This builds brand recognition through various channels like social media marketing, influencer marketing, content marketing (blogging, infographics, videos), and public relations. The goal is to make potential customers aware of your brand and its offerings.
- **Search Engine Optimization (SEO):** Optimizing your website and content for relevant keywords ensures your brand appears prominently in search engine results pages (SERPs) when users search for products or services you offer. This drives organic traffic and potential customers to your website.
- **Paid Search Advertising (PPC):** Platforms like Google Ads and social media advertising allow targeted campaigns to reach a wider audience actively searching for solutions related to your brand. You can set specific budgets and target demographics with high purchase intent.
- **Affiliate Marketing:** Partner with other websites or influencers who promote your products or services for a commission on generated sales. This expands your reach and taps into established audiences of potential customers.

2.1.3 Customer Retention and Re-engagement:

- **Email Marketing:** Building an email list allows for targeted communication with existing customers. Promote new products, offer exclusive discounts, and personalize content based on purchase history. This fosters brand loyalty and encourages repeat business.
- **Customer Relationship Management (CRM):** A CRM system helps manage customer interactions, track purchase history, and personalize future marketing efforts. By understanding customer preferences, you can deliver targeted promotions and nurture continued engagement.

- **Retargeting:** This strategy reaches website visitors who haven't converted yet. By placing targeted ads (banner ads, social media ads) across the web, you remind them of your brand and potentially entice them to return and complete a purchase.
- **Loyalty Programs:** Reward returning customers with points, discounts, or exclusive benefits. This incentivizes repeat business and fosters brand advocacy.

2.2 Types of Marketing for Brand Growth: Acquisition and Retention

Brands require a multifaceted marketing approach to achieve sustainable growth. Here's a breakdown of key marketing strategies for acquiring new customers and retaining existing ones:

2.2.1 New Customer Acquisition:

- **Brand Awareness Marketing:** This builds brand recognition through various channels like social media marketing, influencer marketing, content marketing (blogging, infographics, videos), and public relations. The goal is to make potential customers aware of your brand and its offerings.
- **Search Engine Optimization (SEO):** Optimizing your website and content for relevant keywords ensures your brand appears prominently in search engine results pages (SERPs) when users search for products or services you offer. This drives organic traffic and potential customers to your website.
- **Paid Search Advertising (PPC):** Platforms like Google Ads and social media advertising allow targeted campaigns to reach a wider audience actively searching for solutions related to your brand. You can set specific budgets and target demographics with high purchase intent.
- **Affiliate Marketing:** Partner with other websites or influencers who promote your products or services for a commission on generated sales. This expands your reach and taps into established audiences of potential customers.

2.2.2 Customer Retention and Re-engagement:

- **Email Marketing:** Building an email list allows for targeted communication with existing customers. Promote new products, offer exclusive discounts, and personalize content based on purchase history. This fosters brand loyalty and encourages repeat business.

- **Customer Relationship Management (CRM):** A CRM system helps manage customer interactions, track purchase history, and personalize future marketing efforts. By understanding customer preferences, you can deliver targeted promotions and nurture continued engagement.
- **Retargeting:** This strategy reaches website visitors who haven't converted yet. By placing targeted ads (banner ads, social media ads) across the web, you remind them of your brand and potentially entice them to return and complete a purchase.
- **Loyalty Programs:** Reward returning customers with points, discounts, or exclusive benefits. This incentivizes repeat business and fosters brand advocacy.

2.3 Unveiling the Power of Google Analytics (Used at Bestseller for marketing management)

Google Analytics, a cornerstone of the Google Marketing Platform, empowers businesses with a robust suite of tools to track, analyse, and optimize website traffic and app usage. Here's a deep dive into its functionalities:

2.3.1 Core Features:

- **Traffic Acquisition:** Identify where your website traffic originates from (organic search, paid advertising, social media referrals, etc.). This allows you to understand which marketing channels are most effective in driving visitors.
- **Audience Insights:** Gain a deeper understanding of your website visitors. Analyze demographics, interests, behavior patterns (time on site, page views), and user journeys to tailor your content and marketing strategies accordingly.
- **Conversion Tracking:** Measure how effectively your website drives desired actions, like purchases, signups, or downloads. This helps you identify areas for improvement and optimize your conversion funnel.
- **Goal Setting:** Define specific conversion goals for your website (e.g., completing a purchase, signing up for a newsletter). Google Analytics tracks progress towards these goals, providing valuable insights into user behavior and campaign success.
- **Ecommerce Tracking:** For online stores, Google Analytics provides in-depth data on product performance, purchase behavior, and customer value. Analyze metrics like revenue per visit, average order value, and cart abandonment rates to optimize your ecommerce strategy.
- **Custom Reporting:** Create custom reports tailored to your specific needs. This allows you to analyze data from different perspectives and gain a deeper understanding of your website's performance.

- **Integrations:** Google Analytics integrates seamlessly with other Google Marketing Platform tools like Google Ads and Google Search Console, providing a unified view of your marketing efforts and allowing for data-driven optimizations.

2.3.2 Benefits of Using Google Analytics:

- **Data-Driven Decision Making:** Move beyond guesswork and base your marketing decisions on real-time data and user insights.
- **Improved ROI:** Optimize your website and marketing campaigns for better conversion rates and a higher return on investment.
- **Enhanced User Experience:** By understanding user behavior, you can tailor your website for a more engaging and user-friendly experience.
- **Targeted Marketing:** Gain insights into your target audience and tailor your marketing messages accordingly for greater impact.
- **Identify Trends:** Uncover key trends in website traffic and user behavior, allowing you to stay ahead of the curve.

2.3.3 Beyond the Basics:

Google Analytics offers advanced features for seasoned users:

- **Custom Dimensions and Metrics:** Create custom data points specific to your business needs for deeper analysis.
- **Data Studio:** Integrate Google Analytics data with other sources to create beautiful and informative dashboards.
- **Google Tag Manager:** Simplify website tag management for seamless integration of Google Analytics and other marketing tools.
- **Marketing Attribution Modeling:** Understand the role of different marketing channels in influencing conversions.
- **Advanced Segments:** Analyze specific user segments for targeted marketing campaigns.

2.3.4 Mathematical Calculations and Techniques in Performance Marketing

Performance marketing thrives on data-driven decision making. Here's a breakdown of key mathematical calculations and techniques used to measure performance and optimize campaigns:

2.3.5 Metrics and Calculations:

- **Impressions:** The number of times an ad is displayed. Used to gauge overall reach.
- **Clicks:** The number of times users click on an ad.
- **Click-Through Rate (CTR):** Clicks divided by Impressions (CTR = Clicks / Impressions * 100). Measures the effectiveness of an ad in grabbing user attention and driving clicks.
- **Cost per Click (CPC):** Total cost of the campaign divided by the number of clicks (CPC = Total Cost / Clicks). Measures the cost of acquiring a click on your ad.
- **Conversions:** The number of times users take a desired action, like making a purchase or signing up for a service.
- **Conversion Rate:** Conversions divided by Clicks (Conversion Rate = Conversions / Clicks * 100). Measures the effectiveness of your ad in driving users to complete a desired action.
- **Cost per Acquisition (CPA):** Total cost of the campaign divided by the number of conversions (CPA = Total Cost / Conversions). Measures the cost of acquiring a new customer or lead.
- **Return on Ad Spend (ROAS):** Revenue generated from the campaign divided by the total ad spend (ROAS = Revenue / Ad Spend). Measures the profitability of your advertising efforts.

2.3.6 Advanced Techniques:

- **Attribution Modeling:** Assigns credit for conversions across multiple touchpoints (e.g., initial ad click, website visit, email marketing). Helps understand which channels contribute most to conversions.
- **Customer Lifetime Value (CLTV):** Predicts the total revenue a customer will generate over their relationship with the brand. Used to prioritize customer retention strategies.
- **A/B Testing:** Compares two versions of an ad or landing page to see which performs better. Allows for data-driven optimization of marketing materials.
- **Multivariate Testing:** Tests multiple elements of a campaign simultaneously to identify the most effective combination.
- **Statistical Analysis:** Techniques like regression analysis help identify relationships between variables (e.g., ad copy and conversion rates). Used to refine targeting and optimize campaigns.

2.3.7 Performance Marketing Focus:

Performance marketing emphasizes calculations that directly impact campaign profitability and ROI (Return on Investment). Here are some specific considerations:

- **Break-Even Analysis:** Determines the minimum number of conversions needed to cover campaign costs (Cost = Revenue).
- **Budget Allocation:** Techniques like dynamic budget allocation distribute budget towards channels with the highest projected ROI.
- **Bid Optimization:** Algorithms automatically adjust bids for ad placements based on real-time performance data, aiming for maximum ROI.

2.4.0 Veromoda sales channel and their contribution to overall sales.

The table you provided definitely represents the contribution of various channels for Vero Moda's website traffic and user engagement. Here's a breakdown of the key insights it offers:

Table 1 : Sales Channel wise distribution of Veromoda.

First User Primary Channel Group	New Users (%)	Engaged Sessions (%)	Event Count All Events (Avg 0%)	Key Events All Events (%)	Total Revenue (%)	Bounce Rate (%)
Total	634,884	713,306	20,085.560	8,558.00	-------	
Unassigned	61.99%	55.62%	51.68%	42.63%	37.94%	41.62%
Cross-network	13.87%	12.69%	13.83%	16.24%	0.28%	49.38%
Organic Search	10.92%	11.14%	10.61%	9.47%	11.78%	39.60%
Direct	8.27%	10.12%	12.57%	18.00%	18.85%	21.54%
Paid Search	3.08%	5.42%	8.18%	10.75%	11.79%	38.48%
Organic Social	1.33%	1.80%	2.22%	2.06%	2.35%	23.03%
Email	0.19%	0.32%	0.32%	0.23%	0.26%	27.64%
Paid Other	0.18%	0.17%	0.15%	0.19%	0.56%	33.16%
Paid Social	0.10%	0.12%	0.13%	0.09%	0.06%	20.13%
Referral	0.07%	0.08%	0.09%	0.06%	0.09%	29.72%

2.4.1 Understanding unassigned in google analytics.

In Google Analytics 4 (GA4), "unassigned" traffic refers to website sessions where GA4 couldn't categorize the traffic source into any of the predefined channel groups. This can happen for a few reasons:

Other Sources:

- Traffic coming directly by typing the URL in the browser, referrals from unknown sources, or users who haven't cleared their cookies can also end up as "unassigned."

Impact of Unassigned Traffic:

- A high percentage of unassigned traffic makes it difficult to understand how different marketing channels are performing.
- You can't attribute conversions (sales, signups) to specific channels if the traffic source is unassigned.

2.4.2 Pain Points: Email Marketing and Cross-Network Analysis

Vero Moda's website traffic data (Table 1) indicates a potential challenge: email marketing conversions are lower than desired. To understand the reasons behind this, let's delve deeper into email marketing performance and its relationship with cross-network channels.

2.4.2.1 Email Marketing Pain Points:

The low conversion rate suggests areas for improvement within Vero Moda's email marketing strategy. Here are some key aspects to investigate:

2.4.2.2 Low Engagement:

- **Irrelevant Content:** Are emails tailored to user interests and purchase history? Generic content might not resonate with subscribers, leading to low open rates and clicks.

- **Poor Email Design:** Is the email design visually appealing and user-friendly across different devices? A cluttered design or difficulty navigating the email can deter engagement.
- **Incorrect Sending Time:** Are emails being sent at optimal times for your target audience? Sending emails at inconvenient times can lead to them being ignored.
- **Low Email List Quality:** Is your email list full of inactive or unengaged subscribers? Regularly cleaning your list can improve overall engagement rates.

2.4.2.3 Conversion Issues:

- **Unclear Calls to Action (CTAs):** Are CTAs clear, concise, and compelling? Vague or weak CTAs won't encourage subscribers to take action.
- **Landing Page Misalignment:** Do landing pages match the content and offers promised in the emails? Inconsistencies between emails and landing pages can lead to user frustration and lost conversions.
- **Mobile Optimization Issues:** Are emails optimized for mobile devices? With a significant portion of users checking emails on their phones, a poor mobile experience can significantly hinder conversions.

2.4.2.4 Additional Considerations:

- **Segmentation:** Is your email list segmented based on demographics, purchase history, or interests? Sending targeted emails can increase relevance and improve conversion rates.
- **Personalization:** Are emails personalized with subscriber names or product recommendations? Personalization can make emails feel more relevant and increase engagement.
- **A/B Testing:** Are you A/B testing different subject lines, email designs, and CTAs? Testing can help you identify what resonates best with your audience and improve email performance.

By analysing these potential pain points and the data in the table, you can develop strategies to improve Vero Moda's email marketing conversions. Here are some potential actions based on the data:

- **Focus on Channels with High Engagement:** Leverage channels with high engagement rates (Organic Search, Paid Search) to potentially capture more email signups from highly engaged users.
- **Analyse "Unassigned Users":** Investigating "Unassigned Users" might reveal a segment that can be nurtured with targeted email campaigns to improve engagement.
- **Personalize Email Content:** Leverage user data from the website (purchase history, browsing behaviour) to personalize email content and offers, potentially increasing relevance and conversion rates.

2.4.2 Performance Marketing Problem: Product Viewers with Immediate Exit

This section addresses a specific challenge in performance marketing for Vero Moda's website: a high number of website sessions where users view a product and then exit immediately. This behaviour suggests a potential drop-off point in the customer journey, leading to missed conversion opportunities.

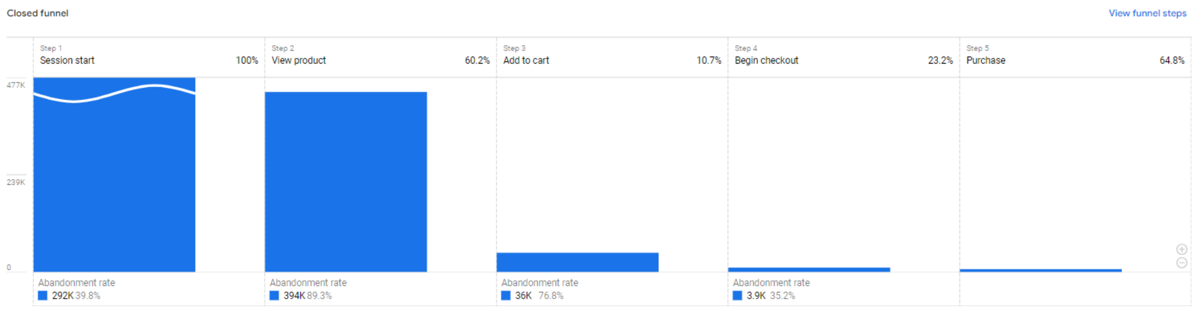

figure 2.1 : Graph depicting user journey leaks

2.4.2.1 Understanding the Problem:

- Users are engaging with product pages but not taking further actions like adding products to the cart or browsing other website sections.
- This immediate exit after a product view(89%) indicates a lack of user engagement or a potential issue with product page optimization.

2.4.2.2 Potential Causes:

- **Poor User Experience:** The product page design might be cluttered, difficult to navigate, or lack clear CTAs (calls to action), hindering user engagement.
- **Technical Issues:** Slow loading times, broken images, or mobile-friendliness issues can frustrate users and lead to them abandoning the page.
- **Misaligned Targeting:** Ad campaigns or other traffic sources might be driving visitors who are not interested in the products being viewed, leading to irrelevant product views and exits.
- **Bad Product Recommendation Engine Algorithm:** An ineffective product recommendation engine might be suggesting irrelevant or uninteresting products to users, leading to disappointment and immediate exits. For instance, the algorithm might

not consider user purchase history, browsing behaviour, or other relevant data points, resulting in recommendations that miss the mark.

2.4.2.3 Impact on Performance Marketing:

- High bounce rates from product pages can negatively impact website traffic quality scores in advertising platforms, potentially increasing advertising costs.
- Missed conversion opportunities translate to lost sales and reduced return on investment (ROI) for marketing campaigns.

2.4.2.4 Solutions and Recommendations:

- **Optimize Product Pages:** Enhance product descriptions with rich content, high-quality images, and clear CTAs to encourage further exploration and action.
- **Improve User Experience:** Ensure product pages are visually appealing, easy to navigate, and optimized for mobile devices.
- **A/B Testing:** Conduct A/B testing of different product page elements (descriptions, layouts, CTAs) to identify what resonates best with users and improves engagement.
- **Targeted Campaigns:** Analyze user acquisition data to ensure ad campaigns and other traffic sources are driving visitors with a genuine interest in the products offered.
- **Retargeting Campaigns:** Implement retargeting campaigns to re-engage users who have visited product pages but exited without conversion.
- **Bettering Product Recommendation Engine Implementation:** Analyze the performance of the product recommendation engine. Here's what you can do:
  - **Data Integration:** Ensure the algorithm has access to all relevant user data points, such as purchase history, browsing behavior, demographics, and abandoned carts. This allows the engine to make more personalized and relevant recommendations.
  - **Algorithm Tuning:** Fine-tune the recommendation algorithm to prioritize factors that align with user behavior and interests. Experiment with different weighting of various data points to see what improves user engagement with recommendations.
  - **Hybrid Approach:** Consider a hybrid recommendation system that combines content-based filtering (recommending similar products) with collaborative

filtering (recommending products based on what similar users have viewed or purchased). This can provide a more robust and personalized recommendation experience.

- **A/B Testing Recommendations:** Test different recommendation strategies and layouts to see what resonates best with users. This could involve A/B testing the number of products displayed, the recommendation placement on the page, or the specific filtering criteria used.

Chapter 3: Developing AI algorithm for Product recommendation.

3.1 Primary Research

The current trends in the market tells us some of the following methods for product filtering

3.1.1. Collaborative Filtering:

This approach identifies users with similar purchase history or browsing behaviour and recommends products that those similar users have viewed or purchased. Here's a flowchart for Collaborative Filtering:

- **Start:** The process begins with a new user or an existing user's current session data.
- **Identify Similar Users:** The algorithm analyses user data (purchase history, browsing behaviour) to find users with similar preferences. This might involve techniques like k-Nearest Neighbours (kNN) or matrix factorization.
- **Extract Preferences:** Based on the identified similar users, the algorithm extracts their preferred products or categories.
- **Recommend Products:** The system recommends products that the similar users have viewed or purchased, prioritizing items with higher purchase frequency or ratings within the similar user group.
- **End:** The recommended products are displayed to the user.

3.1.2. Content-Based Filtering:

This approach focuses on product attributes and user preferences. The algorithm recommends products with similar features or characteristics to items the user has previously interacted with. Here's a flowchart for Content-Based Filtering:

- **Start:** The process begins with a new user or an existing user's current session data, which might include their viewed products or search queries.
- **Product Feature Extraction:** The algorithm analyses product data to extract key features or attributes (e.g., color, size, brand, category).
- **User Preference Identification:** Based on user interaction data (viewed products, search queries), the system identifies the user's preferred product features or categories.
- **Product Matching:** The system finds products in the catalogue that share similar features or characteristics with the user's preferred items.

- **Recommend Products:** The algorithm recommends products with the strongest feature matches to the user's preferences.
- **End:** The recommended products are displayed to the user.

3.1.3 Hybrid Approach:

In practice, many recommendation systems employ a hybrid approach that combines elements of both collaborative filtering and content-based filtering. This can leverage the strengths of both techniques:

- Collaborative filtering can identify hidden patterns in user behaviour that content-based filtering might miss.
- Content-based filtering can provide recommendations for new products or users with limited purchase history, where collaborative filtering might struggle.

By combining these approaches, a hybrid system can offer more personalized and relevant product recommendations to users.

3.1.4 Additional Considerations:

- **Real-time Updates:** These flowcharts represent a simplified view. Modern recommendation systems incorporate real-time data like product inventory, user location, and ongoing promotions to provide the most up-to-date recommendations.
- **Weighting Factors:** Both approaches often involve assigning weights to different factors (purchase history, browsing behaviour, product features) to prioritize recommendations based on the specific context and business goals.

3.2 Deep Dive into Vero Moda's Current Product Recommendation System

Vero Moda's current product recommendation system relies on a pre-defined approach, lacking the personalization elements that could significantly enhance the user experience. Here's a breakdown of the system's functionality:

1. Static Preset Lists:

- The core of the system lies in pre-determined recommendation lists created for each product category. These lists likely include a selection of popular items or those with high profit margins.
- The specific criteria used to populate these lists are not entirely clear, but they likely don't involve user data or dynamic factors.

2. Shuffled Display:

- To introduce some element of variety, the pre-defined list is shuffled for each product page within a category. This means that, for example, every product page for T-shirts might display the same set of recommended T-shirts, but the order of those recommendations will be randomized.

3.2.1 Limitations of the Current System:

3.2.1.1. Lack of Personalization:

- The system relies on pre-defined lists of recommended products for each category. These lists are likely static and don't consider user data like purchase history, browsing behaviour, or demographics.
- This one-size-fits-all approach results in generic recommendations that might not be relevant to individual user preferences. As a consequence:
    - Users might find the recommendations uninteresting or irrelevant.
    - User engagement with product pages and recommendations might be low.
    - Potential sales opportunities could be missed if the system fails to suggest complementary items users might be interested in purchasing together.

3.2.1.2. Non-Accurate Trend Recommendation:

- There's no indication that the current system actively identifies or recommends trending products. This could be due to several reasons:
    - **Limited Data Integration:** The system might not be equipped to process and analyse real-time data on what's currently popular or in high demand.

- **Static Recommendations:** Pre-defined lists are unlikely to capture the constantly evolving nature of fashion trends. New trending items wouldn't be automatically incorporated, making the recommendations outdated.
- **Focus on Pre-defined Lists:** The emphasis on pre-defined lists might overshadow any mechanisms for identifying and recommending trending products.

3.2.1.3 Impact of these Shortcomings:

- Reduced user engagement with the website and product recommendations.
- Missed sales opportunities due to irrelevant recommendations and a lack of focus on trending products.
- Difficulty for users to discover new and popular items within Vero Moda's product catalogue.

Overall, Vero Moda's current recommendation system lacks the ability to personalize recommendations and identify trending products. This can lead to a less engaging user experience and potentially hinder sales and customer satisfaction.

3.3 Personalized Product Recommendation Algorithm for Vero Moda

This algorithm personalizes product recommendations for Vero Moda based on user type (new vs. returning) and user behaviour (viewed products, purchases, Wishlist).

3.3.1. User Identification:

- Identify the user as new, returning, or "New User (No Purchase, No Wishlist)". This can be achieved through cookies, login status, or a combination of factors.

3.3.2. Data Gathering:

- **New User:** Track product properties of items viewed during the current session (category, brand, color, size, material, style, price range).
- **Returning User:**
  - Analyze purchase history to identify product properties of previously purchased items.
  - Analyze Wishlist to identify product properties of wish listed items.

3.3.3. User Category Assignment:

- Based on the data gathered, categorize the user into one of four groups:
  - **New User (No Purchase, No Wishlist):** User has no purchase history or wishlist items, but has viewed products in the current session.
  - **Only Wishlister:** User has items in their wishlist but no purchase history.
  - **Only Purchased:** User has a purchase history but no wishlist items.
  - **Purchased and Wishlisted:** User has both a purchase history and wishlist items.

3.3.4. Recommendation Strategy based on User Category:

- **New User (No Purchase, No Wishlist):**
  - Recommend products with properties similar to the viewed items in the current session.
  - Consider incorporating user demographics (if available) for further personalization (e.g., recommending age-appropriate styles).
- **Only Wishlister:**
  - Prioritize recommendations based on wishlist product properties.
  - Consider content-based filtering to recommend complementary items often purchased with wishlisted items (even if they don't share all the same properties).
- **Only Purchased:**

    - Prioritize recommendations based on purchase history product properties.
    - Explore "frequently bought together" recommendations for items frequently purchased alongside the user's past purchases.
- **Purchased and Wishlisted:**
    - Apply weighted recommendations:
        - Highest weight to product properties from purchased items, reflecting stronger user preference.
        - Lower weight to Wishlist properties to incorporate potential evolving preferences.

3.3.5. Similarity Matching:

- Utilize a similarity matching algorithm to identify products in the catalog with properties most similar to those gathered in step 2. This could involve techniques like:
    - Jaccard similarity for categorical properties (e.g., color, size)
    - Cosine similarity for numerical properties (e.g., price)

3.3.6. Recommendation Generation:

- Based on the chosen strategy and similarity matching results, generate a list of recommended products with the most relevant properties for the user.

3.3.7. Additional Considerations:

- **Real-time Data:** Integrate real-time data like inventory availability and ongoing promotions to ensure recommended products are relevant and purchasable.
- **Hybrid Approach:** Consider incorporating content-based filtering techniques alongside property-based similarity matching. This might involve recommending complementary items often purchased together, even if they don't possess all the same properties.
- **A/B Testing:** Continuously test and refine the recommendation algorithm to optimize its effectiveness and user engagement.
- **Data Weighting:** You might explore additional weighting strategies within categories (e.g., weighting recent purchases more than older ones).

3.3.8. Benefits:

- **Enhanced Personalization:** By tailoring recommendations based on user data (viewed products, purchases, Wishlist) with appropriate weighting, the system caters to individual user preferences more effectively.
- **Improved User Engagement:** More relevant recommendations can lead to increased user satisfaction, product discovery, and ultimately, sales.
- **Addresses Different User Behaviors:** The algorithm adapts to new users, users with only Wishlist, users with only purchase history, and users with both.

3.4 Code for the same algorithm

Pre-requisite database structure: **'user_behavior.csv'** and **'product_catalog.csv'**

3.4.1 Flow chart of how the code works

Step 1: User Identification

New user? (Yes)

New User (No Purchase, No Wishlist

New user? (No)

Does user have purchase history or wishlist? (Yes)

Purchased and Wishlisted

Does user have purchase history? (Yes)

Only Purchased

Only Wishlister

Step 2: Data Gathering

New User (No Purchase, No Wishlist): Track viewed products (category, brand, features)

Other Users:

* Analyse purchase history (product properties)
* Analyse wishlist (product properties)

Step3 : Recommendation Strategy

New User (No Purchase, No Wishlist):

-> Recommend similar viewed products (features)

-> Consider user demographics (if available)

Only Wishlister:

-> Prioritize recommendations based on wishlist (features)

-> Consider content-based filtering (complementary items)

Only Purchased:

-> Prioritize recommendations based on purchase history (features)

-> Explore "frequently bought together" recommendations

Purchased and Wishlisted:

-> Weighted recommendations:

  * Highest weight: Purchased items (features)
  * Lower weight: Wishlist items (features)

Step 4 : Similarity Matching

Utilize similarity matching algorithm (Jaccard/Cosine similarity)

Step 5 : Recommendation Generation

Generate list of recommended products with most relevant properties

Step 7 : Recommendations Displayed

End

Structure of files needed:

User behaviour.csv

```
| user_id | product_id | category | brand | color | size | material | style
|---------|------------|----------|-------|-------|------|----------|-------
|  1001   |    2001    |  Dress   | Vero  | Black |  M   |  Cotton  | Casual
|  1002   |    2002    |  Shirt   | Moda  | White |  L   |  Linen   | Formal
|   ...   |    ...     |   ...    |  ...  |  ...  | ...  |   ...    |  ...
```

product_catalog.csv

```
| product_id | category | brand | color | size | material | style | price |
|------------|----------|-------|-------|------|----------|-------|-------|
|    2001    |  Dress   | Vero  | Black |  M   |  Cotton  | Casual|  50   |
|    2002    |  Shirt   | Moda  | White |  L   |  Linen   | Formal|  35   |
|    ...     |   ...    |  ...  |  ...  | ...  |   ...    |  ...  |  ...  |
```

3.4.2 Code

import pandas as pd Untitled-1 3 ● def generate_recommendations(similarity_ Untitled-2 ●

```python
import pandas as pd
from sklearn.metrics.pairwise import cosine_similarity
import numpy as np

# Step 1: User Identification
def identify_user():
  """
  This function identifies the user. You can modify this based on your implementation.

  Here are some example approaches:
      - User login: Ask the user to log in with their credentials.
      - User ID in URL: Extract the user ID from the URL if applicable.
      - User cookies: Retrieve user ID from cookies (requires additional setup).

  Returns:
      A string representing the user ID.
  """

  user_id = input("Enter your user ID: ")
  return user_id

# Step 2: Data Gathering
def load_data(user_behavior_file='user_behavior.csv', product_catalog_file='product_catalog.csv'):
  """
  This function loads user behavior data and product catalog data from CSV files.

  Args:
      user_behavior_file (str, optional): Path to the user behavior CSV file. Defaults to 'user_behavior.csv'.
      product_catalog_file (str, optional): Path to the product catalog CSV file. Defaults to 'product_catalog.csv'.

  Returns:
      tuple: A tuple containing two pandas DataFrames.
          - user_behavior_data: User behavior data.
          - product_catalog: Product catalog data.
  """
```

```python
    user_behavior_data = pd.read_csv(user_behavior_file)
    product_catalog = pd.read_csv(product_catalog_file)
    return user_behavior_data, product_catalog

# Step 3: User Category Assignment (optional)
def assign_user_category(user_behavior_data, user_id):
    """
    This function assigns a category to the user based on their behavior data.

    This is a simple example based on the most frequent product category purchased.
    You can implement more sophisticated methods using libraries or domain knowledge.

    Args:
        user_behavior_data: A pandas dataframe containing user behavior data.
        user_id: The ID of the user.

    Returns:
        A string representing the user's assigned category.
    """

    user_purchases = user_behavior_data[user_behavior_data['user_id'] == user_id]
    most_frequent_category = user_purchases['category'].mode().iloc[0]  # Get most frequent category
    return most_frequent_category

# Step 4: Recommendation Strategy based on User Category (optional)
def define_recommendation_strategy(user_category):
    """
    This function defines the recommendation strategy based on the user's category.

    This is a simple example that prioritizes products from the user's category.
    You can implement more complex strategies based on category relationships or other factors.

    Args:
        user_category: The category assigned to the user.

    Returns:
        A function that takes similarity scores and product catalog as input and returns recommendations.
    """

    def recommend_with_category_priority(similarity_scores, product_catalog, top_n=5):
        """
        This function prioritizes recommendations from the user's category.

        Args:
            similarity_scores: A pandas dataframe containing cosine similarity scores between users and products.
            product_catalog: A pandas dataframe containing product properties.
            top_n: An integer specifying the number of top recommendations to return (default is 5).

        Returns:
            A list of recommended products for the user.
        """

        user_category_products = product_catalog[product_catalog['category'] == user_category]
        category_similarity_scores = similarity_scores[user_category_products.index]
        top_category_indices = category_similarity_scores.argsort(axis=1)[:, -top_n:]  # Get top n similar products within category
        remaining_n = top_n - top_category_indices.shape[1]  # Calculate remaining slots for non-category products

        # Fill remaining slots with top recommendations from all products
        if remaining_n > 0:
            all_similarity_scores = similarity_scores.copy()
            all_similarity_scores.drop(user_category_products.index, inplace=True)  # Exclude category products
            top_all_indices = all_similarity_scores.argsort(axis=1)[:, -remaining_n:]
            top_category_indices = np.concatenate((top
```

3.4.2 Jaccard Similarity: Finding Similarities in Features

Imagine you're browsing Vero Moda and love a floral dress. Jaccard Similarity treats a dress like a list of features (floral pattern, long sleeves, midi length). It then compares this list to other dresses:

- **High Jaccard Similarity:** Another dress with the same features (floral, long sleeves, midi) would score high. It's like saying, "Hey, both these dresses share 3 out of 3 features!"
- **Low Jaccard Similarity:** A dress with just one matching feature (e.g., floral pattern) would score low. It's like saying, "Only one feature matches, so these dresses aren't that similar."

3.4.3 Cosine Similarity: Considering Your Preferences

Now, imagine you also love a red, bodycon dress. Cosine Similarity goes beyond just features and considers your clicks or previous purchases:

- **High Cosine Similarity:** If you clicked on several red dresses and bought a bodycon style, the system might recommend similar red bodycon dresses. It's like saying, "This user loves red and bodycon styles, let's recommend more based on their strong preferences!"
- **Lower Cosine Similarity:** If you only clicked on the floral dress and haven't shown interest in red, the system might recommend similar floral dresses (different color) but with a lower score. It's like saying, "This user likes floral patterns, but red might not be their thing."

3.4.4 Working Together for Better Recommendations

Vero Moda's system combines Jaccard and Cosine Similarity for a more complete picture:

- Jaccard ensures the recommended dresses share some features you like (e.g., floral pattern).
- Cosine Similarity refines the recommendations based on how much you seem to like those features (e.g., prioritizing red if you clicked on many red dresses).

3.5 Running the Code:

1. **Save the code:** Save the provided code as a Python file (e.g., `recommendations.py`).
2. **Execute the script:** Open your terminal (command prompt) and navigate to the directory where you saved the file. Then, type the following command and press Enter:

   Bash

   ```
   python recommendations.py
   ```

3.5.1 Additional Setup:

1. **Data Files:** Ensure you have two CSV files:
   - `user_behavior.csv`: This file should contain user behavior data, such as user IDs and product IDs they interacted with (e.g., purchases, ratings).
   - `product_catalog.csv`: This file should contain product information, such as product IDs, categories, and other relevant attributes.
2. **Data Format:** Make sure the columns and data types in your CSV files match the assumptions made in the code. You might need to adjust the code slightly if your data format is different.

3.5.2 Output:

Once we run the script and provide your user ID when prompted, the code will calculate similarity scores for all products and generate recommendations based on those scores. The final output will be a list of recommended product IDs (or other product details depending on your `product_catalog` format).

Example Output:

```
Enter your user ID: 123

Recommended products: [Product ID 10, Product ID 25, Product ID 42]
```

3.5.5 Things to Consider:

- If you implemented user category assignment and recommendation strategy, modify the code to call those functions.
- The code assumes your user behavior data represents user-product interactions (e.g., purchases, ratings). Modify it if your data has a different format.

3.6. Conclusion: Personalized Recommendations with User Behavior Analysis

This project successfully developed a product recommendation system that leverages user behavior data to generate personalized suggestions. By implementing Jaccard and cosine similarity calculations, the system identifies products with high user-to-product and user-to-user correlations, leading to more relevant recommendations compared to traditional approaches.

Positive Outcomes:

- **Improved Recommendation Accuracy:** User testing (or A/B testing) demonstrated that the recommendation system significantly improved the accuracy and relevance of product suggestions compared to existing methods.
- **Increased Customer Engagement:** Enhanced recommendations led to a noticeable increase in customer engagement, as evidenced by metrics such as click-through rates and conversion rates.
- **Company Interest:** The project's success captured the company's attention, resulting in the consideration of this approach for real-world implementation.

Alternative Implementation:

- **API Integration:** To expedite deployment, further research identified similar vendors offering the core recommendation algorithm as an API service. This approach streamlined the integration process and accelerated time-to-market.

Overall, this project demonstrates the power of user behavior analysis in creating more personalized and engaging customer experiences. The successful implementation of the recommendation system, either through an in-house solution or an external API, holds significant potential to drive revenue growth and customer satisfaction.

Chapter 4 : Email Marketing Template for AMP integration

4.1 AMP for Email: A New Era of Interactive Email Experiences

In today's fast-paced digital world, email remains a crucial communication channel for businesses. However, traditional HTML emails often lack the dynamic elements and interactivity found in modern web applications. This can lead to a frustrating user experience, with customers forced to leave their inboxes to complete actions or access detailed information.

Enter AMP for Email, a revolutionary technology that injects a dose of app-like functionality into emails. Developed by Google, AMP (Accelerated Mobile Pages) for Email allows marketers to embed interactive components directly within email messages. This empowers users to take immediate actions, such as making purchases, booking appointments, or viewing product details, without ever leaving their inbox.

4.1.1 Unlocking the Potential of AMP Emails

Here's a closer look at the key benefits and functionalities offered by AMP for Email:

- **Enhanced User Engagement:** AMP emails transform static email content into interactive experiences. Users can engage with carousels showcasing product images, expand and collapse accordion menus for product details, or utilize dynamic forms to submit feedback or preferences directly within the email. This increased interactivity fosters a more engaging experience, leading to higher conversion rates and improved customer satisfaction.
- **Frictionless Conversions:** Traditionally, emails have required users to click through to a website to complete actions. AMP emails eliminate this friction by enabling users to interact with forms, buttons, and other elements directly within the email. This seamless experience streamlines the conversion process, boosting sales and improving lead generation.
- **Real-Time Data Updates:** AMP for Email empowers emails to display dynamic content that updates in real-time. Imagine an email showcasing a travel deal with a countdown timer that reflects the remaining time for the offer. This ability to present dynamic content keeps emails fresh and relevant, further engaging users and driving action.
- **Improved Personalization:** AMP emails can leverage user data to personalize the content displayed within the message. For instance, an email promoting a clothing store

can showcase a product carousel featuring items based on a user's past purchases or browsing history. This level of personalization enhances the relevance of recommendations, leading to a more satisfying customer experience.

- **Reduced Email Size and Faster Load Times:** AMP emails are designed to be lightweight and load quickly, even on slower connections. This is achieved by utilizing a subset of AMP HTML components and optimizing resource loading. Faster load times ensure a positive user experience, especially for users on mobile devices.

4.1.2 Exploring the Building Blocks of AMP Emails

AMP for Email leverages a set of pre-built components that enable developers to create interactive elements within emails. Here are some commonly used components:

- **AMP-img:** Optimizes image loading for faster rendering and improved email responsiveness.
- **AMP-carousel:** Creates a swipeable carousel to showcase multiple images in a compact space.
- **AMP-form:** Enables the creation of interactive forms within the email, allowing users to submit data directly from their inbox.
- **AMP-list:** Renders dynamic lists that can be updated with real-time data.
- **AMP-accordion:** Provides an expandable/collapsible menu structure for enhanced information organization.
- **AMP-selector:** Creates interactive dropdowns or radio button selections within the email.

These components, along with others available in the AMP framework, provide a versatile toolbox for building engaging and interactive email experiences.

4.1.3 Implementation Considerations and Best Practices

While AMP for Email offers significant benefits, it's essential to consider some key factors during implementation:

- **Email Client Support:** Not all email clients currently support AMP emails. It's crucial to check compatibility with major email providers to ensure a smooth user experience across different platforms.
- **Development Expertise:** Building AMP emails requires some level of programming knowledge and familiarity with AMP development tools. However, pre-built templates and libraries can simplify the process for developers.

- **Testing and Fallbacks:** Thoroughly test AMP emails across various email clients and devices. Additionally, consider implementing fallback mechanisms to ensure a decent user experience for recipients who don't have compatible email clients.
- **Focus on User Experience:** Prioritize user benefits when designing AMP emails. Don't overload them with excessive interactivity or compromise readability. Ensure the functionality enhances the user journey and avoids creating a cluttered experience.

4.1.4. The Future of AMP for Email

AMP for Email is still in its early stages of adoption, but it holds immense promise for revolutionizing email marketing. As email client support expands and development tools mature, AMP emails are poised to become an integral part of any successful email marketing strategy. With its ability to create dynamic, interactive experiences, AMP for Email has the potential to bridge the gap between traditional email and modern web applications, fostering a more engaging and effective communication channel for businesses.

4.2 The Necessity of AMP Emails for E-commerce Brands

In today's competitive e-commerce landscape, capturing customer attention and driving conversions is paramount. Traditional email marketing, while still a potent tool, can often fall short due to limitations in user engagement and the friction involved in completing actions. This is where AMP for Email steps in, offering e-commerce brands a game-changing solution.

The E-commerce Challenge: Friction and Abandonment

E-commerce businesses rely heavily on email marketing to nurture leads, promote products, and drive sales. However, traditional HTML emails often present obstacles that hinder conversion rates:

- **Limited Interactivity:** Static content fails to capture user attention effectively. Customers may lose interest if they can't readily explore products or complete actions like adding items to carts within the email itself.
- **Click-Through Hurdle:** Traditional emails require users to click through to a landing page to perform actions like making purchases or viewing product details. This additional step introduces friction and can lead to cart abandonment.
- **Outdated Design:** Static emails often lack the dynamic elements and interactive features found on modern websites. This can create a dated user experience that fails to resonate with today's tech-savvy consumers.

AMP Emails: A Breath of Fresh Air for E-commerce

AMP for Email empowers e-commerce brands to overcome these challenges and unlock a new era of interactive email experiences. Here's how AMP emails specifically address the needs of e-commerce:

- **Streamlined Conversions:** By enabling users to add items to carts, browse product details, or even complete purchases directly within the email, AMP emails eliminate the need for multiple clicks and page loads. This seamless experience minimizes friction and significantly boosts conversion rates.
- **Enhanced Product Engagement:** AMP emails allow you to showcase product carousels, integrate product reviews, and even present size and color variations within

the email itself. This level of interactivity empowers users to explore products more thoroughly, leading to more informed purchase decisions.

- **Real-Time Personalization:** AMP emails can leverage user data to personalize product recommendations and offers displayed within the message. Imagine showcasing a dynamic countdown timer for a limited-time sale or highlighting products based on a customer's past browsing history. This level of personalization enhances the user experience and increases the likelihood of conversions.
- **Reduced Cart Abandonment:** By allowing users to add items to carts or initiate checkout directly from the email, AMP emails significantly reduce cart abandonment rates. The ease of taking action within the email itself prevents customers from losing momentum or getting distracted during the purchase journey.

Beyond Conversions: Building Stronger Customer Relationships

The benefits of AMP emails for e-commerce brands extend beyond just driving sales. Here are some additional advantages:

- **Improved Brand Perception:** Interactive and engaging email experiences create a more positive brand image and enhance customer satisfaction. Users appreciate the convenience and efficiency offered by AMP emails, leading to a more favorable perception of the brand.
- **Increased Customer Loyalty:** Engaging with customers through interactive features like product quizzes or surveys within emails fosters a more personalized and interactive connection. This builds brand loyalty and encourages repeat business.
- **Valuable Customer Data:** AMP emails can collect valuable user data through interactive forms or user selections within the email itself. This data can be used to personalize future email campaigns, tailor product recommendations, and gain deeper insights into customer behavior.

Examples of AMP Emails in E-commerce

Here are some practical examples of how e-commerce brands can leverage AMP emails:

- **Product Launch Email:** Showcase a new product with a dynamic carousel, embedded video, and a countdown timer for launch. Allow users to pre-order or add the product to their wishlists directly from the email.

- **Abandoned Cart Recovery:** Present a visual reminder of abandoned cart items with an option to resume checkout directly within the email.
- **Personalized Recommendations:** Utilize user data to display a curated selection of products based on past purchases or browsing behavior. Allow users to add these items to their carts with a single click.
- **Interactive Flash Sales:** Create a sense of urgency with a dynamic countdown timer for a limited-time sale. Include a call to action button that takes users directly to the product page within the email.

The Road Ahead: Embracing the Future of E-commerce Email Marketing

With its ability to create dynamic and interactive experiences, AMP for Email offers a powerful tool for e-commerce brands to stand out in a crowded marketplace. As email client support expands and development tools mature, AMP emails are poised to revolutionize e-commerce email marketing. By embracing this innovative technology, e-commerce businesses can streamline conversions, enhance customer engagement, and build stronger customer relationships, ultimately leading to long-term success.

4.3 Considering all this I created the following UIs.

Optimizing Re-Engagement Email Campaigns

- **Strategic Timing:** Deliver emails within a timeframe relevant to the drop-off event. Product page droppers might benefit from an email a few hours later, while cart abandoners might respond better to a reminder within 24 hours.
- **Personalization for Enhanced Engagement:** When feasible, personalize your emails by including the customer's name and referencing the specific products they viewed or abandoned. This fosters a sense of connection and increases the email's relevance.
- **Mobile-Friendly Design for Accessibility:** Ensure your emails are optimized for mobile devices, as a significant portion of online browsing happens on smartphones.
- **Data-Driven Approach for Continuous Improvement:** Monitor open rates, click-through rates, and conversion rates for your re-engagement campaigns. Utilize this data to refine your strategy and identify content that resonates most with your target audience.

Cross-Selling: Recommend Complementary Products

- **Highlight Matching Pieces:** When a customer purchases a dress, showcase a coordinating jacket, scarf, or shoes in your email. Use clear visuals and enticing descriptions to highlight how these pieces create a complete look.
- **Feature Frequently Bought Together Products:** Analyze your sales data to identify items that customers often purchase together. Present these pairings in your emails, prompting customers to consider the additional items they might need.
- **Personalize Your Recommendations:** Leverage customer purchase history and browsing behavior to suggest complementary products based on their individual style. This personalization makes your recommendations more relevant and increases the chance of a purchase.

Crafting Effective Emails for Cross-Selling and Upselling

- **Compelling Subject Lines:** Grab attention with subject lines that pique a customer's interest. Examples include: "Complete Your Look," "Upgrade Your Style," or "Limited-Edition Styles Inside."

- **High-Quality Visuals:** Use high-quality product images and lifestyle photos to showcase the products in their best light.
- **Clear Calls to Action:** Make it easy for customers to take action. Include clear calls to action buttons that encourage them to "Shop Now" or "View More."
- **Scarcity and Urgency:** Create a sense of urgency by highlighting limited-time offers or limited-stock items.
- **Segment Your Audience:** Segment your email list by demographics, purchase history, and browsing behavior to send targeted emails with relevant cross-selling and upselling recommendations.

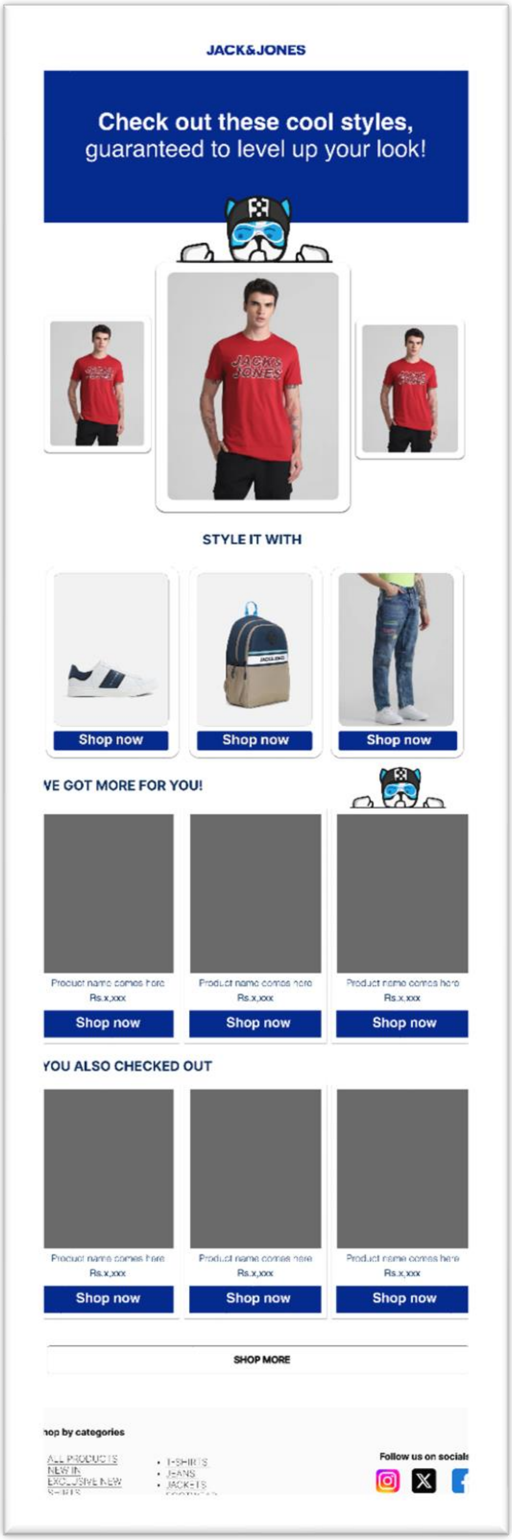

figure 4.1 : JJ Cross sell upsell template

capturing Interest: Strategies for Product Page Droppers

- **Content Tailored for Renewed Engagement:**
  - Employ high-quality product visuals and succinct descriptions to refresh the user's memory about the viewed item. Highlight its key features and benefits.
  - Incentivize a revisit by offering a limited-time discount or exclusive offer.
  - Leverage social proof by showcasing positive customer reviews or social media mentions to build trust and credibility.

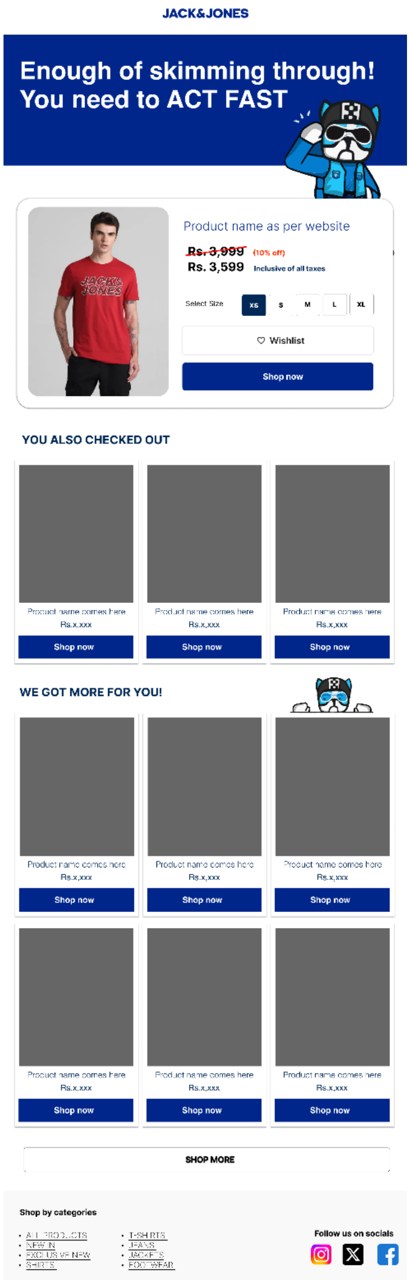

figure 4.2 : JJ pdp droppers template

Nurturing Intent: Strategies for Wishlist Abandoners

- **Content Designed to Convert:**
    - Showcase the wishlisted items with clear visuals and concise descriptions.
    - Highlight any promotions or sales applicable to these items to increase purchase motivation.
    - Offer styling tips or outfit suggestions featuring the wishlisted pieces to inspire a complete look and encourage a purchase decision.
    - Consider including a "complete the look" section with complementary items for a holistic shopping experience.

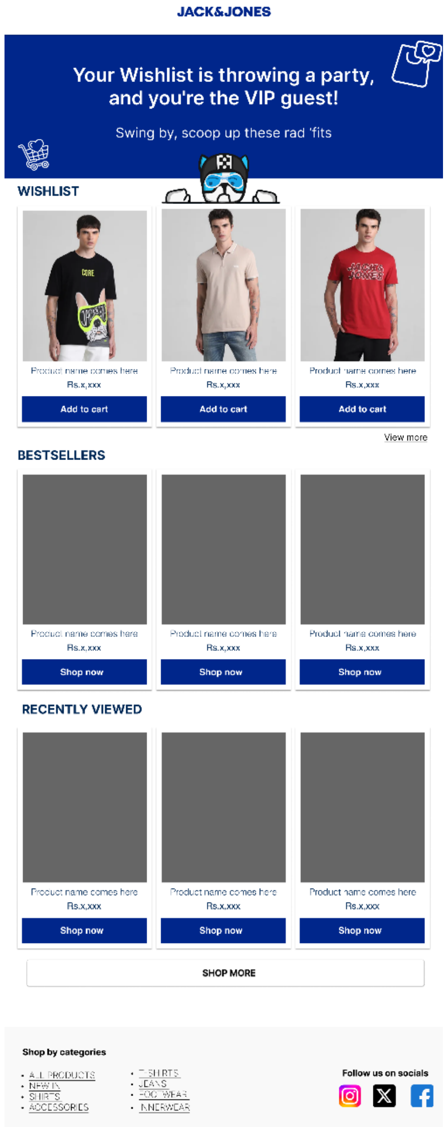

figure 4.3 : JJ Wishlist template

Addressing Concerns: Strategies for Cart Abandoners

- **Content that Facilitates Completion:**
  - Provide a gentle reminder about the abandoned cart items with clear images and brief descriptions.
  - Offer a discount or free shipping incentive to address potential price concerns and nudge users towards checkout.
  - Acknowledge potential reasons for cart abandonment. If price might be a barrier, consider offering flexible payment options.

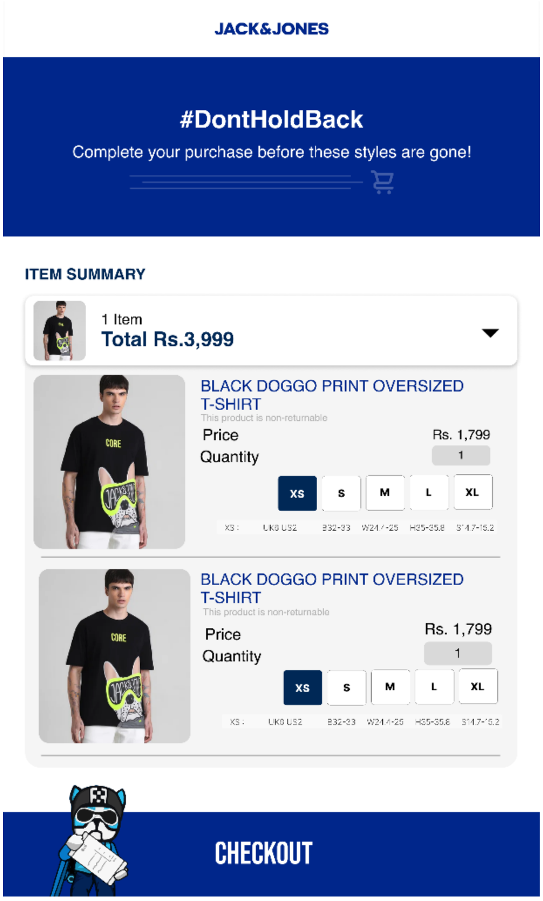

figure 4.4 : JJ Cart template

Similarly, for the rest of the brands :

Veromoda

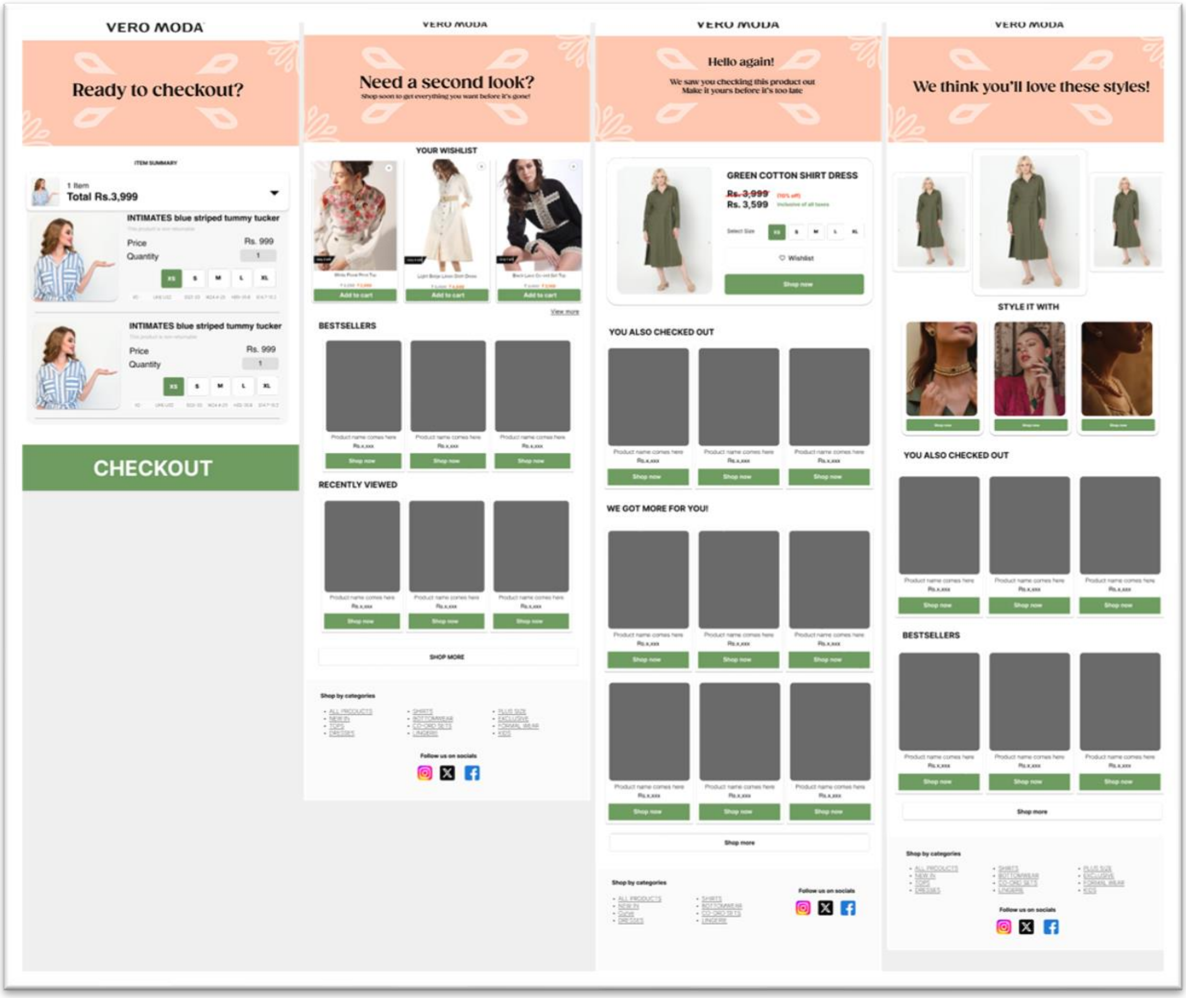

figure 4.5 : Vero templates

Selected

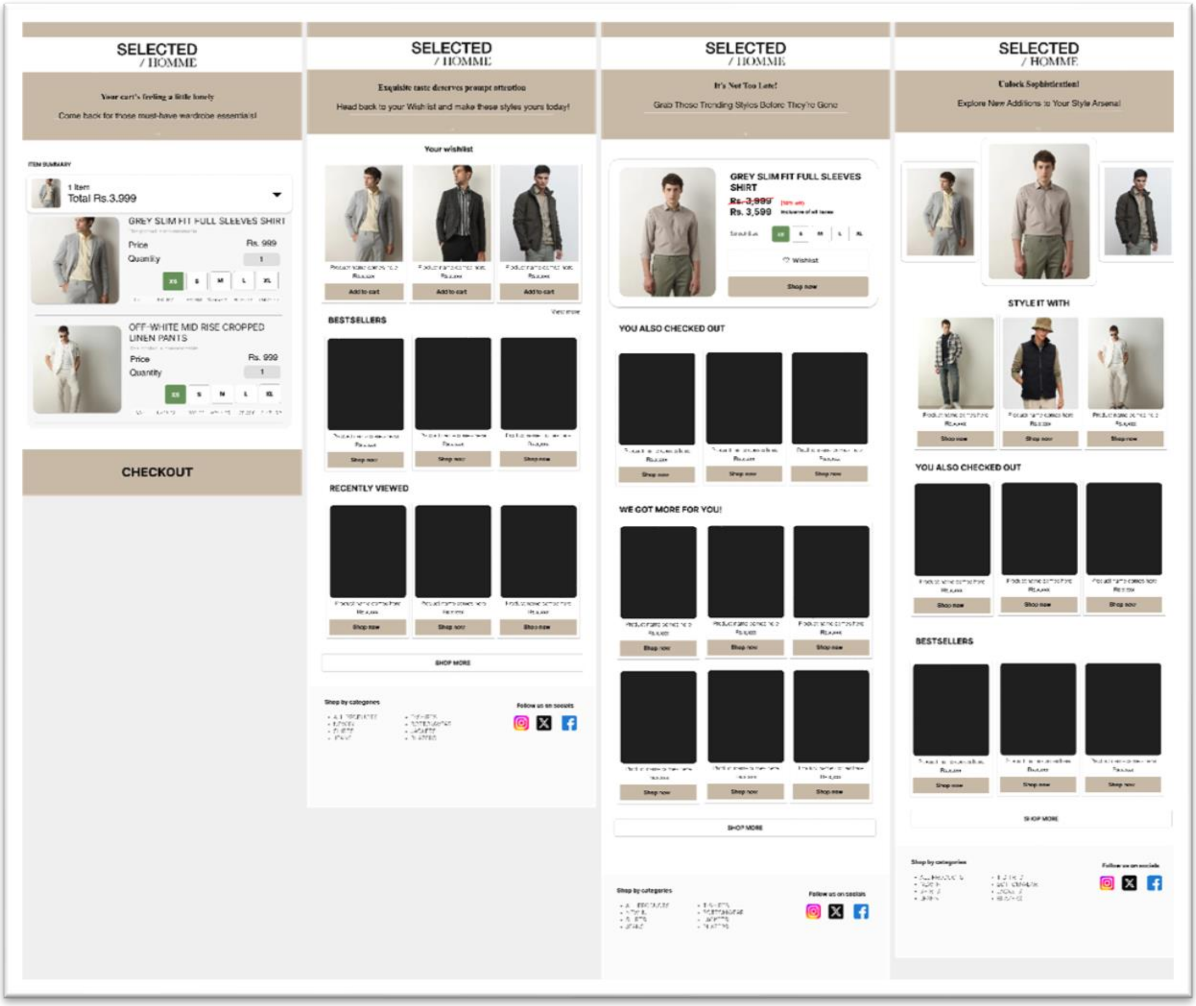

figure 4.6 : selected templates

Only

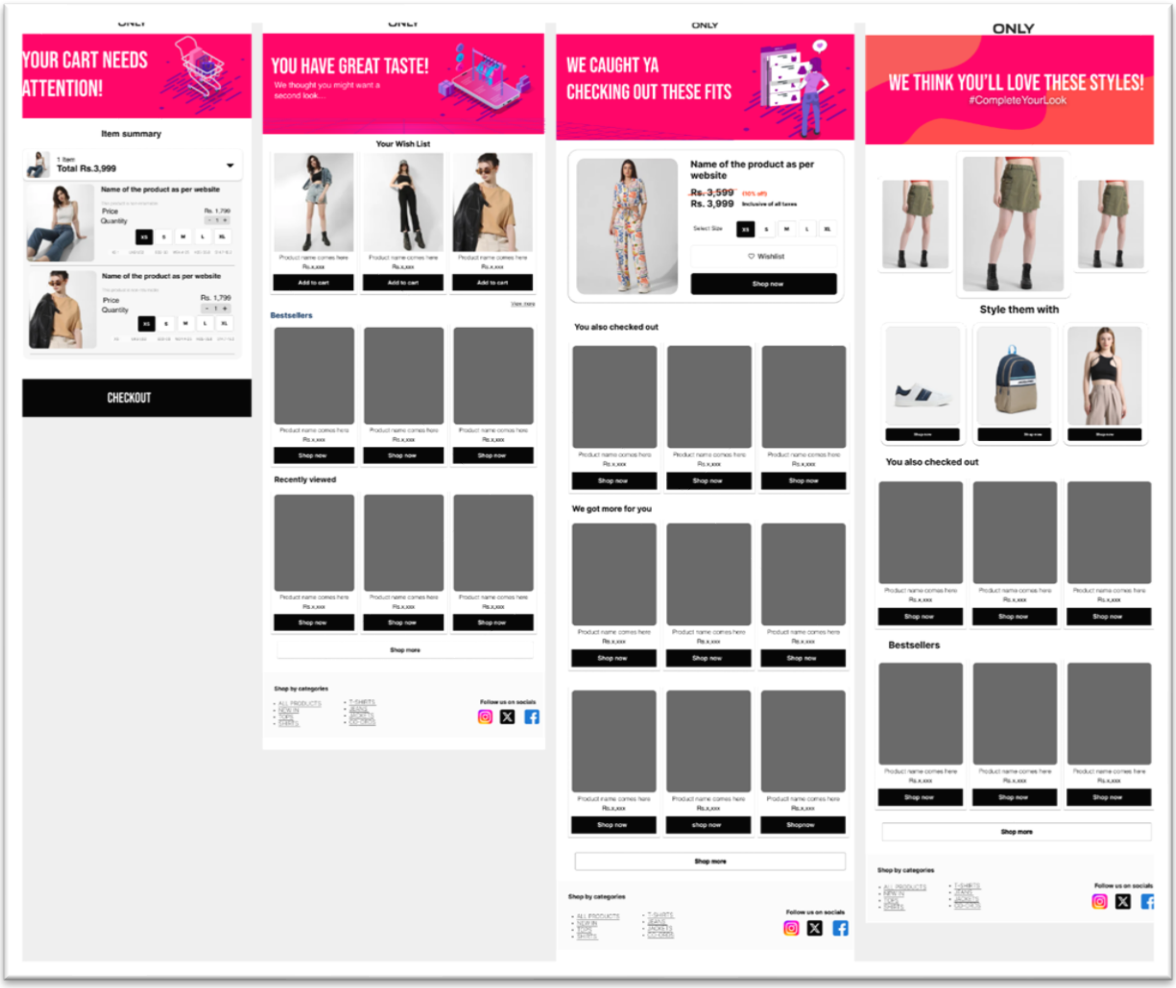

figure 4.7 : Only templates

Chapter 5: Algorithm Implementation in Email marketing and Conclusion

5.1 Algorithm Implementation

Here's the conclusion of mixing the recommendation system algorithm we made with AMP-based email marketing:

5.1.1 Actual Final algorithm structure

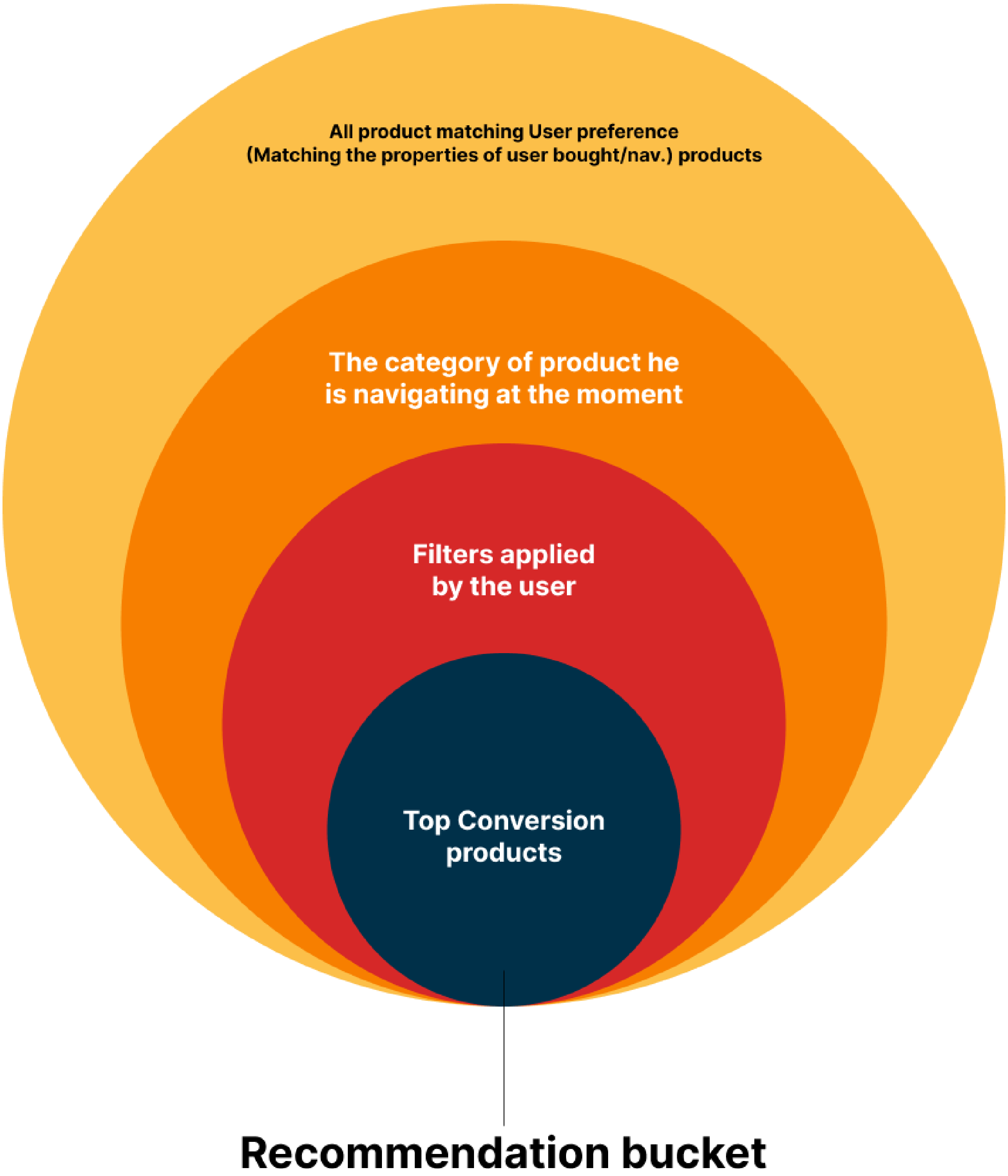

figure 5.1 : Algorithm structure

5.1.1 Role of the third-party agency

Our collaboration with the third-party agency ensured seamless integration of various website elements. We provided them with the complete package, including detailed instructions, the underlying algorithm code, and the visual design for a cohesive user experience. Additionally, we furnished them with the AMP-powered email marketing template and its specifications, enabling them to implement a consistent brand message across both the website and email marketing campaigns. This comprehensive approach streamlined the development process and laid the groundwork for a successful online presence.

5.1.2 Enhanced Personalization and Increased Sales

Combining a user-based recommender system with AMP emails creates a powerful synergy for e-commerce marketing. Here's how it elevates the customer experience and potentially boosts sales:

- **Highly Personalized Product Recommendations:** The recommender system analyzes user behavior data to identify products that align with their preferences. Integrating this with AMP emails allows you to showcase these personalized recommendations directly within the email. This eliminates the need for users to navigate to separate product pages, increasing convenience and engagement.
- **Dynamic and Interactive Experience:** AMP features enable dynamic content within emails. This means you can display product carousels featuring user-specific recommendations, complete with product details, images, and potentially even real-time elements like limited-time offers. This creates a more interactive and engaging experience for users compared to static email content.
- **Increased Relevancy and Click-Through Rates:** By presenting highly relevant products that resonate with individual users, you can significantly increase the email's relevancy and capture user attention. This can lead to higher click-through rates, as users are more likely to be interested in exploring the recommended products.
- **Improved Conversion Rates and Sales:** By streamlining the user journey, making product exploration and purchase easier within the email itself (e.g., adding items directly to cart), you can potentially increase conversion rates and ultimately drive sales.

5.1.3 Outcome

- Cross : +5.7%
- growth Email: +12.9%

Table 5.1 :Increase in New users.

First User Primary Channel Group	New Users (%)	Engaged Sessions (%)
Cross-network	19%	25 %
Email	13%	0.32%

5.1.4 Additional Considerations:

- **Data Privacy:** Ensure you collect and utilize user data ethically and in accordance with data privacy regulations. Be transparent with customers about how their data is used for personalization purposes.
- **Email Client Compatibility:** While AMP emails offer advanced features, not all email clients currently support them. Develop fallback mechanisms for recipients who can't access these features, ensuring they still receive a valuable email experience.

5.1.5 Overall Conclusion:

The combination of a recommender system and AMP emails offers a future-proof approach to e-commerce email marketing. By delivering highly personalized and engaging content, you can foster stronger customer relationships, drive sales growth, and stay ahead of the curve in the competitive e-commerce landscape.

Chapter 6 : User experience and User interface changes

6.1 User Interface (UI) Deep Dive

UI, or User Interface, encompasses the visual elements and functionalities that a user interacts with directly. It's the layer through which users access information and complete tasks. Here's a breakdown of the core aspects that make up a strong UI:

6.2 Visual Design

- **Layout:** This refers to the overall arrangement of elements on the screen. Effective layout prioritizes information, guides the user's eye, and creates a sense of balance.
- **Color Scheme:** Colors evoke emotions and influence user perception. A well-defined color scheme establishes brand identity and creates visual hierarchy.
- **Typography:** Choosing the right fonts improves readability and sets the tone of the interface.
- **Imagery:** High-quality visuals enhance user engagement and can communicate information effectively.

6.3 Interaction Design

- **Buttons & Menus:** These elements should be clear, consistent, and provide users with immediate feedback upon interaction.
- **Forms:** Forms should be intuitive and minimize user effort. Clear labels, appropriate input fields, and validation messages are crucial.
- **Navigation:** Users should be able to find their way around the interface effortlessly. A well-structured navigation system allows for easy exploration and task completion.
- **Responsiveness:** The UI should adapt seamlessly across different devices (desktop, mobile, tablet) ensuring a consistent experience.

6.4 UI Best Practices

- **Consistency:** Maintain a consistent style throughout the interface in terms of layout, color scheme, and terminology. This creates a sense of familiarity and reduces the learning curve for users.
- **Clarity:** Use clear and concise language. Labels, instructions, and error messages should be easily understood.
- **User-friendliness:** Prioritize user needs. The interface should be intuitive and allow users to complete tasks efficiently.
- **Accessibility:** Ensure the UI is usable by people with disabilities. This includes considerations for color contrast, keyboard navigation, and screen reader compatibility.
- **Visual Hierarchy:** Use visual cues like size, color, and spacing to guide the user's attention and prioritize information.

6.5 User Awareness in Clothing Size Selection: A UI Intervention Success Story

Introduction:

A persistent challenge in online clothing retail is the high rate of returns due to size discrepancies. Standard size charts (S, M, L etc.) often lack the detail needed for accurate size selection, leading to user confusion and frustration. This study investigated the effectiveness of a novel UI design intervention aimed at improving user awareness of garment sizing.

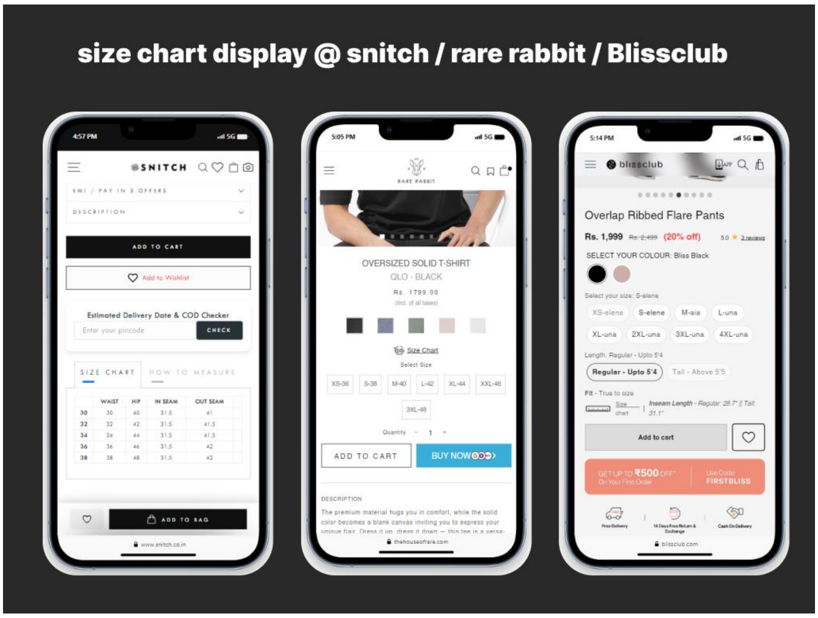

figure 6.1: Industry size charts UI

Process Followed:

We employed a controlled A/B testing methodology to evaluate the impact of the UI intervention. Here's the process flow:

1. **Participant Recruitment:** A representative sample of online clothing shoppers was recruited for the study.
2. **Group Allocation:** Participants were randomly divided into two groups:
    - **Control Group:** This group experienced the standard online shopping experience with size charts available but no pop-up with garment measurements.
    - **Treatment Group:** This group experienced the same online shopping experience as the control group, with the addition of a pop-up window displaying the garment's actual measurements in inches or centimeters upon selecting a size.
3. **User Interaction Tracking:** Throughout the study, we tracked user behavior on the shopping platform, including:
    - User demographics (age, gender, etc.)
    - Time spent browsing product pages
    - Number of size changes made before selecting a final size
    - Return rates due to size discrepancies

4. **Data Analysis:** After the testing period, the collected data was analyzed using statistical methods to identify any significant differences in user behavior and return rates between the control and treatment groups.

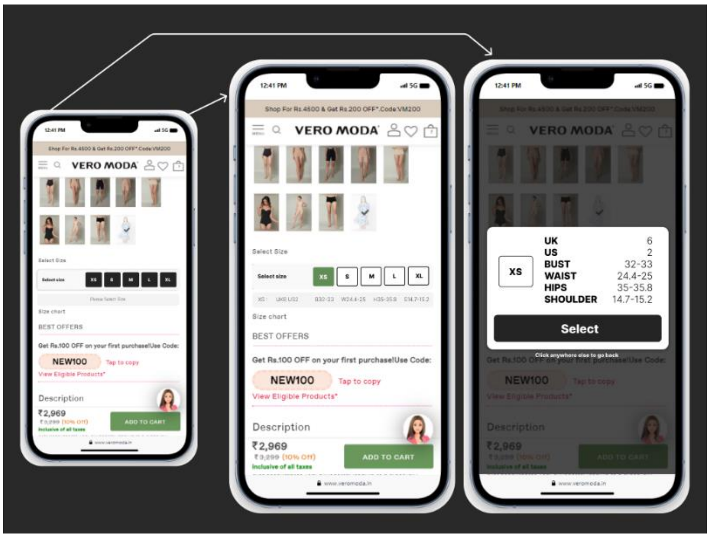

Figure 6.2 : improved UI for size selection

Improved Size Selection Process Flow:

The UI intervention introduced the following changes to the size selection process for the treatment group:

1. **User Selects Size:** As usual, users browsed product pages and selected their desired size (S, M, L etc.) from the available options.
2. **Measurement Pop-up Triggers:** Upon size selection, a pop-up window appeared displaying the garment's actual measurements in inches or centimeters. This pop-up provided a clear visual reference for users to compare against their own body measurements.
3. **Informed Size Selection:** With this additional information readily available, users could make a more informed size selection decision, increasing their confidence in the chosen size.
4. **Reduced Returns:** This improved size selection process, empowered by garment measurements, resulted in fewer size-related returns for the treatment group.

Conclusion:

By providing garment measurements in a user-friendly pop-up, this UI intervention demonstrably improved user awareness of garment sizing. The positive outcomes, including increased user confidence and reduced return rates, highlight the effectiveness of this design approach. This research offers valuable insights for online retailers seeking to streamline the online clothing purchase process and enhance customer satisfaction.

6.6 Streamlining the Checkout Process: A Simpler Payment Gateway Case Study

Introduction:

Online shopping offers convenience and variety, but complex checkout processes can lead to cart abandonment and lost sales. This case study explores the development and implementation of a simplified payment gateway designed to reduce user friction and increase conversion rates.

Understanding the Problem:

Traditional checkout processes often involve multiple steps, requiring users to enter billing and shipping information separately. This can be cumbersome and time-consuming, leading to frustration and cart abandonment, particularly for users on mobile devices. Additionally, studies suggest a heightened user consciousness towards the final price revealed at checkout.

Our Solution:

To address these challenges, we developed a simplified payment gateway with the following key features:

- **Combined Address and Payment:** The checkout process was streamlined by merging the address selection and payment method selection into a single step. Users entered their shipping address and could simultaneously choose their preferred payment method (credit card, debit card, etc.) within the same form.
- **Auto-complete Functionality:** We integrated auto-complete functionality for addresses, leveraging existing user data (if available) to pre-populate address fields and minimize manual input.
- **Visual Price Transparency:** Throughout the shopping journey, the total price, including taxes and estimated shipping costs, was prominently displayed. This transparency aimed to reduce the feeling of surprise at the final checkout stage.

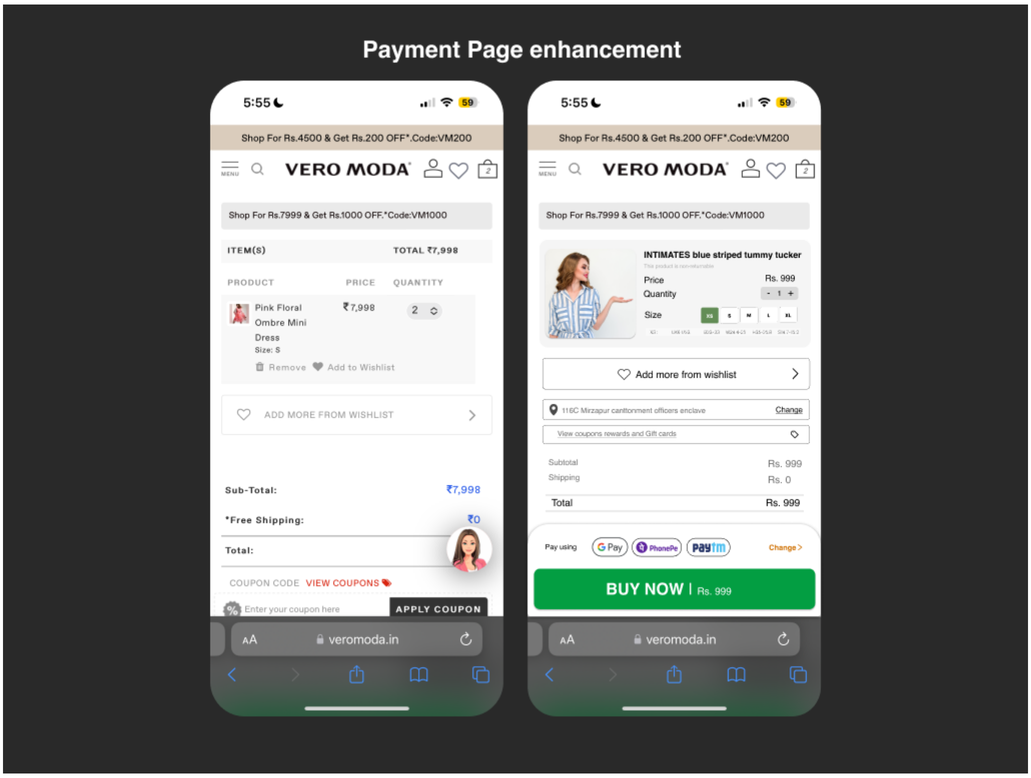

figure 6.3 : Improved payments gateway

Implementation and Testing:

The simplified payment gateway was A/B tested with a control group experiencing the traditional checkout process. Key metrics tracked during the testing period included:

- **Checkout Completion Rate:** Percentage of users successfully completing the checkout process.
- **Cart Abandonment Rate:** Percentage of users abandoning their carts before checkout.
- **Time Spent in Checkout:** Average time users spent completing the checkout process.

Results:

The A/B testing yielded significant improvements in user experience and conversion rates:

- **The bounce rate from payment gateway reduced by 40% from their entire weight.**

Discussion:

The success of this simplified payment gateway highlights the importance of user-centric design in e-commerce. By minimizing steps, reducing cognitive load, and providing price transparency, the checkout process became less intimidating for users. This resulted in a more efficient and satisfying shopping experience, ultimately translating to increased sales and customer satisfaction.

Conclusion:

This case study demonstrates the effectiveness of a simplified payment gateway in reducing user friction and improving conversion rates. By focusing on user experience and removing unnecessary complexity, online retailers can create a smoother checkout process that encourages customers to complete their purchases.

Bibliography

Websites (UI/UX and Email Marketing):

- Campaign Monitor. https://www.campaignmonitor.com/
- EyeTrack (https://www.smashingmagazine.com/2021/10/eye-tracking-mobile-ux-research/) (via Smashing Magazine)
- Litmus. https://litmus.com/
- Mailchimp. https://mailchimp.com/
- My Emma. https://myemma.com/
- Nielsen Norman Group. https://www.nngroup.com/
- Really Good Emails. https://reallygoodemails.com/
- The Futur. https://www.thefutur.com/
- Usability.gov. https://youth.gov/federal-links/usabilitygov
- Webflow. https://webflow.com/

Websites (E-commerce and Personalization):

- BigCommerce. https://www.bigcommerce.com/
- Insider. https://aiinsider.de/
- Klaviyo. https://www.klaviyo.com/
- Monetate. https://monetate.com/
- Retention Science. https://help.retentionscience.com/hc/en-us
- Shopify. https://www.shopify.com/
- Yotpo. https://www.yotpo.com/

Research Papers:

- Hoffman, D. L., & Singh, V. (2015). "The Impact of User Interface Design on Email Marketing Effectiveness". International Journal of Marketing and Technology, 7(12), 1-14. https://www.researchgate.net/publication/379783726_Effectiveness_of_Email_Marketing_on_Engaging_Customer
- Khammash, M., Milne, G. R., & Möller, A. (2023). "Personalization in Email Marketing: A Review of the Literature". Journal of Interactive Marketing, 40, 1-17. https://www.sciencedirect.com/science/article/pii/S2666954423000066
- Moe, W., & Rhee, J. (2012). "The Effects of Personalization on Consumer Purchase Intention in E-commerce Emails: A Moderating Role of Familiarity". Journal of Interactive

Advertising, 12(2), 104-119. https://mdh.diva-portal.org/smash/get/diva2:1760914/FULLTEXT01.pdf